

Slip Ring Induction Rotor

1. Single squirrel cage induction rotors-it has only one winding in the rotor.
2. Double squirrel cage induction rotors-it has two windings in the rotor.

Construction of a Slip Ring Induction Motor

The rotor of the slip ring motor is actually wound for three phase windings connected in star formation. These windings are placed in insulated rotor slots and the ends are brought out and connected to the three slip rings mounted on the shaft.

Brushes mounted on the slip rings connect the rotor windings to external resistance for the purpose of starting.

This motor is started with full additional resistance in the rotor circuit to ensure less starting current and more starting torque. The external resistance is reduced gradually and the required speed is obtained by short-circuiting the three slip rings. The rotor winding will have the same number of poles found in the stator. This motor is also known as "wound rotor induction motor".

Difference between a Squirrel Cage Induction Motor and a Slip-Ring Induction Motor:

Squirrel Cage Induction Motor	Slip Ring Induction Motor
1. Simple in construction	Construction is complicated because rotor is also wound with insulated wire.
2. Low cost	The cost is high
3. Operates at high efficiency	Efficiency is not high
4. There is no chance of sparks and hence it is explosion proof.	Sparking risk is there because of slip rings and the brushes riding on them.
5. Requires the least maintenance.	Maintenance is trouble some
6. Starting arrangements are simple.	Requires costly starters.
7. Starting torque is less.	High starting torque.
8. Speed control is not easily possible.	Speed can be changed easily.

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